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7CCSMPRJ MSc Project

SAFETY AND SECURITY INTERPLAY (VARIANT 2)

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Table of Contents

1. Introduction	3
1.1 Problem Description, Background, and Motivation	3
1.2 Project Aim and Objectives.....	3
1.3 Technical Specifications.....	4
1.4 Methodology.....	4
2. Background Material	5
2.1 Literature Review	5
2.2 Existing Approaches by Others	6
3. Project Schedule	8
Sprint 1	8
Sprint 2	8
Sprint 3	8
Sprint 4	9
4. References.....	9

1. Introduction

1.1 Problem Description, Background, and Motivation

Multi-layered system architecture is an approach to system architecture design where a system can be split into different layers that represent different purposes, interfaces, and rules regarding the system and its overall requirements. Implementing this approach can allow for components and subsystems to be arranged accordingly for easier understanding, verification, maintenance, and design.

The number of autonomous vehicles has risen significantly in recent years, but do the security properties (e.g. sensor data authenticity) on one layer compose to guarantee the safety properties (integrity of obstacle detection mechanisms) at a different layer of vehicle system architecture? How do architectural constructs, such as redundancy, hierarchical decomposition, and shared resources, upon which these systems may be based, govern whether this cross-layer composition holds?

Autonomous vehicles are marketed as the future of transportation and as a luxury, but how can they truly be successful and reliable when most of the public is too hesitant to trust them with complete control over driving?

Given existing cases about malfunctioning autonomous vehicles, such as some Waymo self-driving vehicles failing to sense oncoming pedestrians [1], it's understandable that the public would not want to be completely left in the hands of an autonomous vehicle.

It raises questions about the designs of these existing systems and how factors such as redundancy, shared resources, and hierarchical decomposition are considered in these existing designs.

If the cross-layer composition between safety and security cannot be guaranteed, then it could allow external malicious sources to exploit these systems. An example of a malicious source would be the potential risk that a malicious message could hijack a self-driving car and cause it to hit pedestrians (risking safety AND security). This can appear in the form of a physical attacker holding a clear sign that can misdirect the vehicle [2] (similar to SQL Injection Attacks on Automatic Number Plate Recognition software [3]).

1.2 Project Aim and Objectives

The aim of this project is to develop a robust system architecture that remains safe and secure-by-design, regardless of different architectural constructs that would be applied to ensure an efficient autonomous driving system

The first objective is to ensure that architectural patterns that maintain safety and security properties are present and only benefit the overall system

The second objective is to examine redundancy, shared resources, and hierarchical decomposition within the system.

The third objective is to design the system architecture using a chosen Unified Modelling Language (UML) tool.

The fourth objective is to investigate the chosen architectural patterns (Redundancy, Shared Resources, and Hierarchical Decomposition) to produce architectural designs based on each pattern.

The fifth objective is to perform Compositional Verification, where each component is verified and tested individually to ensure the whole system works correctly.

The sixth objective is to test each design pattern through specific safety and security scenarios that model relationships between both for certain elements across different layers

1.3 Technical Specifications

For the specification of this project, it is decided that the system architecture designs are to be produced in Eclipse Papyrus supporting the generation of Java code, for the system architecture at layers of the multi-layered system model, and therefore the internal relationships and constructs of these designs. Eclipse Papyrus is an industrial-grade open-source model-based engineering tool [4]. Upon initial usage, it works as a UML (Unified Modelling Language) tool that allows the user to use code to implement different components and variables into the design interface, as well as regular drag-and-drop that every other UML tool allows for.

Due to the nature of Formal Modelling requiring precise and correct semantics with no ambiguity in its mathematical representations, the use of Java code can help facilitate a correct-by-semantics system design.

1.4 Methodology

In order to carry out this project, I am choosing to adopt an Agile methodology and divide this workload into “sprints”.

Agile Methodology is a project methodology, catered towards software development, that uses an iterative and incremental approach to decide upcoming work at various points throughout the project during set meetings between “sprints”, which are the different work periods of the project. This methodology also implements concepts such as “Kanban” and “Adaptive System Development”.

In the first sprint, the main activities revolved around learning and researching the topic at hand, as well as understanding the problem to be solved and forming the framework of the solution to be produced.

In the second sprint, this very preliminary report has been written which encapsulates the current progress of the project, as well as acting as the initiation of the implementation phase of the project, where development of the base system architecture begins. The entire development of the project’s systems will take place within Eclipse Papyrus.

In the third sprint, the implementation phase will continue and will aim to conclude by the end of the sprint. Along with this, preparations will be made for carrying out Compositional Verification for each design, verifying whether the designs comply with the safety and security scenarios and allow for the underlying properties to be preserved, and to begin drafting the final project report. Report findings throughout the implementation and verification phases. The most progress will take place during this period and will allow for more flexibility based on any changes as per the feedback from the supervisor.

In the fourth sprint, the project will aim to conclude with the final project report, along with the video presentation being submitted.

2. Background Material

2.1 Literature Review

Literature	Brief Overview
Yellman, T.W. (2006, January). ‘ <i>Redundancy in Designs</i> ’	Identifies two types of related failures that undermine redundancy: cascading failures (where one triggers another) and common-external-cause failures (where a single event affects multiple components). Uses jet engines as examples.
Jean, X., Gatti, M., Faura, D., Pautet, L., Robert, T. (2013, October). ‘ <i>A software approach for managing shared resources in multicore IMA systems</i> ’	Addresses timing unpredictability in safety-critical multicore avionic systems caused by shared hardware resource contention. The authors propose constraining resource usage to predefined "usage domains" where

	processor behaviour remains deterministic and analysable for certification.
Krishnamachari, R.S., Papalambros, P. (1995, August). <i>'Hierarchical Decomposition Synthesis in Optimal Systems Design'</i>	Formalises how to decompose large system optimisation problems into hierarchical subproblems by systematically selecting objective functions for each level. This overall approach makes decomposition rigorous.
Cofer, D., Gacek, A., Miller, S., Whalen, M. W., LaValley, B., Sha, L. (2012, April). <i>'Compositional verification of architectural models'</i> . In NASA Formal Methods Symposium, Berlin, Heidelberg: Springer Berlin Heidelberg.	Presents a methodology for designing and verifying complex cyber-physical systems using formal methods that scale to large, real-world systems

One key concept of this project is the architectural construct of Redundancy, where additional, independent components or subsystems hold duplicate functions/instructions to currently in-use components and subsystems to act as a backup in case of failure. In the context of Autonomous Vehicles, this could be spare sensors, actuators, or communication paths in case of failure to ensure the vehicle remains operational.

Another key concept of this project is the architectural construct of Shared Resources, which are standalone system components, such as sensors, actuators, processors, communication paths, or memory, that are simultaneously used by multiple subsystems or processes. However, this can strain the components as they become single points of potential failure, leading to a risk of multiple processes being compromised if a single component were to go down. Though, this construct can be cost-effective and generally efficient in design.

Another key concept of this project is the architectural construct of Hierarchical Decomposition, which is the process of breaking a complex system into smaller, structured layers that represent different aspects and responsibilities of the larger system itself. This approach can aid design, analysis, verification, and risk assessment through arranging and simplifying parts of the system into different levels of abstraction, from a higher level to a lower level.

A key method used to validate the system architecture designs will be Compositional Verification, which is a technique used in architectural models to individually test and verify components to see if they function properly by themselves, with other components, and within the subsystems they're placed in.

2.2 Existing Approaches by Others

Literature	Brief Overview
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Chattopadhyay, A., Lam, K-Y., Tavva, Y. (2020, June). <i>'Autonomous Vehicle: Security by Design'</i>	Addresses safety and security concerns with autonomous vehicles and explores possibilities of integration, along with potential challenges. Includes a focus on threat, adversarial, operational, and trust modelling along with business mindsets in order to propose safety-by-design and secure-by-design Autonomous Vehicle (AV) systems
Khoka, S. (2024, October). <i>'Integrating Safety and Security in Autonomous Vehicles: A Comprehensive Review'</i>	Addresses safety and security concerns with autonomous vehicles and explores possibilities of integration, along with potential challenges. Includes a focus on Artificial Intelligence, Machine Learning, and Digital Twin systems along with an explicit consideration for safety and security interplay.

One existing approach to this field would be the study “Autonomous Vehicle: Security by Design” (2020) (listed in the table above). In the study, the authors delved into the risks that autonomous vehicles face and their levels of autonomy, the different security issues a Cyber-Physical System (CPS) has and what similar issues an AV system could have as a result of being a derivative of a CPS, the possible approach that ensures security-by-design through a traditional but modified methodology to outline weaknesses and countermeasures, and the following implementation, operation and management that would be needed to bring the study’s proposed approach to life and maintained for years to come. The author also carried out simulated cyberattacks that tests the proposed AV model and demonstrates how security-by-design principles mitigate them.

Some main criticisms of this study that differs from my proposed aims and objectives is that the authors don’t utilise formal modelling to demonstrate their proposed safe-by-design and secure-by-design AV system model, along with a lack of different architectural constructs that their AV can be based around (such as redundancy).

Another existing approach to this field would be the study “Integrating Safety and Security in Autonomous Vehicles: A Comprehensive Review” (2024) (listed in the table above). In this study, the author delved into the current methodologies for integrating safety and security in AVs whilst emphasising the interdependent nature of the two aspects and also considers the advancements in Artificial Intelligence (AI) and Machine Learning (ML) as relevant factors to consider for real-time threat detection and mitigation. They also assesses the trends, challenges, and future directions of AV development, where they believe that AI and ML development, Vehicle-to-Everything (V2X) communication, and use of Digital Twins are major trends, the maintenance of strong safety and security being a challenge in development in terms of financial and computational, and future directions revolving around enhancing all of the prior technologies and concepts mentioned.

Although the study does mention redundancy as a viable method in strengthening the relationship between safety and security (primarily through the mention of Digital Twin systems), a main criticism of this study that differs from my project is that formal methods are also not present here either.

3. Project Schedule

	March	April	May	June	July	August
Sprint 1	5 th March	2 nd April				
Sprint 2		3 rd April 30 th April				
Sprint 3			1 st May		2 nd July	
Sprint 4					3 rd July	6 th August

This table illustrates the project schedule with dates listed

Sprint 1

- Set up GitHub repository with Preliminary Report document created
- Fix minor details from 2nd meeting slides
- Prepare slides for 3rd meeting (and apply feedback after the meeting)
- Extend literature review

Sprint 2

- Start drafting Preliminary Report
- Begin designing the base system architecture

Sprint 3

- Prepare slides for 4th meeting (and apply feedback after the meeting)
- Prepare for 1st individual meeting
- Finish designing the base system architecture
- Produce the variant designs for each of the architectural constructs (off of the base system architecture)
- Conduct Compositional Verification on variant designs and report findings
- Begin drafting Final Project Report

Sprint 4

- Prepare for 2nd individual meeting
- Complete and submit Final Project Report
- Complete and submit Video Presentation

4. References

[1]: Reuters, “US opens probe after a Waymo self-driving car hit child near a school”, Al Jazeera, January 29th, 2026. [Online]. Available at: <https://www.aljazeera.com/economy/2026/1/29/us-opens-probe-after-a-waymo-self-driving-car-hit-a-child-near-a-school>

[2]: L. Burbano, D. Ortiz, Q. Sun, S. Yang, H. Tu, C. Xie, Y. Cao, A.A. Cardenas, “CHAI: Command Hijacking against embodied AI”, arXiv, September 30th, 2025. [Online], p. 3. Available at: <https://arxiv.org/pdf/2510.00181>

[3]: RoadieRich, “Speed camera SQL Injection”, Reddit, July 29th, 2013. [Online]. Available at: https://www.reddit.com/r/geek/comments/1j9tn3/speed_camera_sql_injection/

[4]: Eclipse Foundation, “Eclipse Papyrus Modelling Environment”, Eclipse Papyrus homepage. Available at: <https://eclipse.dev/papyrus/>